AJAYKRISHNAN E S

Email : aj66@ucsb.edu
Website: ajaykrishnan-es.github.io

EDUCATION

Ph.D. in Computer Science — 2023 - current
University of California, Santa Barbara — CGPA - 4/4
Advisor: Prof. Daniel Lokshtanov
Topics: Graph Algorithms, Algorithmic Graph Theory

BS-MS (Specialisation in Mathematics) — 2018 - 2023
Indian Institute of Science Education and Research, Pune — CGPA - 9.2/10
Thesis Title: Exact Exponential Algorithms & Knot-Free Vertex Deletion

AWARDS AND RECOGNITION

Recipient of **Chuck Dana Research Accelerator Award** at UC Santa Barbara. — 2026
Awarded $8,000; sole recipient from a competitive pool of UCSB CS graduate students.

Co-author of the paper that received the **Mihai Pătrașcu Best Paper Award** at SOSA. — 2026
Title: A Quasi-Polynomial Time Algorithm for 3-Coloring Circle Graphs.

Recipient of the **Outstanding Teaching Assistant** award at UC Santa Barbara. — 2025
Nominated by graduating seniors; awarded to one TA per department annually.

Recipient of UCSB CS **Summer Academic Fellowship**. — 2024
Awarded $10,000 to support independent research over the summer.

Recipient of the **Xytel Best Thesis Award** in Mathematics at IISER Pune. — 2023
Awarded to one graduating student per discipline each year.

Graduated with distinction from IISER Pune. — 2023
An honor reserved for students with a CGPA above 9 (approximately the top 5%).

Recipient of the **Innovation in Science Pursuit for Inspired Research** scholarship. — 2018
Five-year fellowship from the Government of India for pursuing higher studies in science.

VISITS AND WORKSHOPS

SoCal Theory Day at University of California San Diego — Feb 2026

Institute of Mathematical Sciences, Chennai. — Summer 2025
Host: Prof. Saket Saurabh.
Topic: Parameterized Algorithms.

Workshop on Hardness of Approximation in P at DIMACS, Rutgers University — July 2025

Fusing Theory and Practice of Graph Algorithms at ICERM, Brown University — Feb 2025

Extroverted Sublinear Algorithms at Simons Institute — June 2024

Randomized Methods for Approximation and Parameterized Algorithms. — Dec 2022

Algorithms - To Improve, or Not to Improve, That's the Question! — Jan 2022

SELECT PUBLICATION

Note: In theoretical computer science, authors are listed in alphabetical order. Among my publications at venues which use ranked authorship, those marked with (⋆) indicate equal contribution from all authors.

Parameterized Approximation of Rectangle Stabbing.
Huairui Chu, **Ajaykrishnan E S**, Daniel Lokshtanov, Anikait Mundhra, Thomas Schiber, Xiaoyang Xu, Jie Xue. *ESA 2026*

A Quasi-Polynomial Time Algorithm for 3-Coloring Circle Graphs. ***(Best Paper Award)***
Ajaykrishnan E S, Robert Ganian, Daniel Lokshtanov, Vaishali Surianarayanan. *SOSA 2026*

Fast Hypertree Decompositions via Linear Programming: Fractional and Generalized. (⋆)
Vaishali Surianarayanan, Anikait Mundhra, **Ajaykrishnan E S**, Daniel Lokshtanov. *SIGMOD 2025*

Beyond Exact Fairness: Envy-Free Incomplete Connected Fair Division.
Ajaykrishnan E S, Daniel Lokshtanov. *FSTTCS 2025*

UNDER SUBMISSION

A Trichotomy for Tree-Independence of Induced-Minor-Closed Classes. *Submitted - SODA*
Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov.

Algorithms for Min-Color Path with Bounded Color-Degree. *Submitted - FOCS*
Ajaykrishnan E S, Daniel Lokshtanov, Saket Saurabh, Jie Xue.

(Treewidth, Clique)-Boundedness and Poly-logarithmic Tree-Independence. *Submitted - IJM*
Maria Chudnovsky, **Ajaykrishnan E S**, Daniel Lokshtanov.

Induced Minors and Coarse Tree Decompositions. *Submitted - JCTB*
Maria Chudnovsky, Julien Codsi, **Ajaykrishnan E S**, Daniel Lokshtanov.

Computing Join Trees for Join Queries with Inequalities. *Submitted - PODS*
Ajaykrishnan E S, Daniel Lokshtanov, Anikait Mundhra, Vaishali Surianarayanan, Nikolaos Tziavelis.

SERVICE

Conference Reviewing:

FOCS 2024, ITCS 2025, COCOON 2025, ESA 2025, STACS 2026, SODA 2026, IWOCA 2026, MFCS 2026.

Mentorship Roles:

UC Santa Barbara, Early Research Scholars Program; a research program that seeks to give students, *particularly those who are traditionally under-served* in computing the foundational knowledge and skills for engaging in research in the discipline.

- Mentored a team of three undergraduate students. 2024-2025
  Project Title: Teaching Assistant Assignment.
- Mentoring a team of four undergraduate students. 2025-2026
  Project Title: Low Distortion Metric Embedding.

Teaching Experience:

- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II
  With Prof. Subhash Suri; Fall 2023, Spring 2024, Winter 2025.
- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II
  With Prof. Eric Vigoda; Winter 2024.
- Teaching Assistant at UCSB for CMPSC 130B - Data Structures and Algorithms II
  With Prof. Daniel Lokshtanov; Fall 2024.
- Teaching Assistant at UCSB for CMPSC 134 - Randomized Algorithms
  With Prof. Eric Vigoda; Spring 2025.
- Teaching Assistant at UCSB for CMPSC 40 - Foundations of Computer Science
  With Dr. Vaishali Surianarayanan; Summer 2024.
- Teaching Assistant at IISER-Pune for MTH 323 - Graph Theory
  With Prof. Soumen Maity; Fall 2022.